

Helix OTA — New Constitution Tech Adoption Checklist (2026-07-09)

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Context

The constitution submodule working tree is at e60cbde2 (59 commits ahead of what the parent repo's git tree currently records). This checklist is grounded in direct inspection of the constitution's new anchors + tooling dirs, cross-checked against what THIS project (helix_ota) actually has wired, per §11.4.6 (verify, don't assume).

FACT #0 — submodule pointer itself is stale (§11.4.26 violation).

```
$ git -C constitution rev-parse HEAD
e60cbde2e33e7d0542a937331763f6be7e739c9a
$ git ls-tree HEAD constitution
160000 commit fc4e9b8ec30909ba7e4aaeb4ab31f9ec15ab5fde
constitution
```

The parent repo's tracked submodule commit (fc4e9b8e) is BEHIND the submodule working tree (e60cbde2). Per §11.4.26 step 7, the consuming project's pointer MUST be bumped in the same commit as any cascade work — this is currently out of sync and is itself the first item to fix.

New anchors summarized (what each REQUIRES)

Anchor	Requires
§11.4.176	Multi-track work-division: (A) exactly-once work-item/logical- group claim registry so two

Anchor	Requires
§11.4.182	tracks never claim the same item; (B) capability-aware, deadlock-proof device-lock (all-or-nothing, non-blocking, TTL-reap). Only binds when the project runs ≥ 2 concurrent tracks/worktrees on shared resources. Every agent/subagent dispatch description AND every operator-facing work-stream reference MUST start with (T<N>/<branch> - <alias>). Mechanically enforced by a PreToolUse guard hook (guard-track-branch-label.sh) that BLOCKS any Agent/Task dispatch whose description doesn't match $\wedge (T[0-9]+/[^\wedge])+\wedge$. Requires the hook wired in .claude/settings.json + a convention doc.
§11.4.183	Every track/work-stream MUST maximize multi-agent usage (subagent-driven dev, independent review agents, parallel background streams) AND apply the entire constitution, nothing skipped — bounded by §12.6 memory + §11.4.58 agent cap + genuine parallelizability. No new tooling; it's a working-mode discipline, verified via propagation gate only.
§11.4.184	sonar-scanner CLI MUST be installed AND durably PATH-discoverable (.bashrc/.zshrc), proven GREEN by sonarqube_install_check.sh exit 0. Mandates tooling availability, NOT a scan on every build

Anchor	Requires
	(distinct from the repealed §11.4.166 Semgrep mandate). Rootless podman only (§11.4.161). No scope of work/release is “done” until it receives manual QA-team confirmation as the FINAL step, in addition to (never instead of) every automated gate. The autonomous-loop release-tag terminal condition (§11.4.126) now REQUIRES this human confirmation; the agent must hand off and WAIT (while progressing other non-blocked work in parallel), never self-certify. Any project with more than one representation of the same tracked data (this project: docs/workable_items.db + Issues.md/Fixed.md + summaries + md/html/pdf/docx exports) MUST run a deterministic cross-document consistency gate (DEDUP/TIMELINE/CROSS-DOC/INTEGRITY/STRUCTURAL checks) BEFORE any export render, doc/DB verify, or doc-set commit — never an after-the-fact human audit. Engine = constitution/scripts/doc_integrity (Go, depth-1 reusable, consumer supplies a checkset.yaml). Before scaling parallel subagents/processes, check <code>ulimit -u (soft+hard) + live thread count (ps -L -u "\$USER" \\\ wc -l);</code> thread exhaustion (EAGAIN/failed to create new OS thread) is a §12 host-safety event requiring unconditional yield/
§11.4.185	
§11.4.186	
§12.12	

Anchor	Requires
	serialize, distinct from the §12.6 memory ceiling.

Tooling inventory (constitution submodule — all PRESENT, verified by 1s)

All of the following exist under constitution/scripts/ and constitution/submodules/ — real files, not aspirational:

- sonarqube/ — sonarqube_install_check.sh, sonarqube_run_scan.sh, sonarqube_container.sh, sonarqube_lib.sh, compose/
- doc_integrity/ — Go module (cmd/, internal/, wire/), README.md, NOTES_FOR_CONDUCTOR.md; ships wire/CM-DOC-INTEGRITY-VALIDATION.gate.sh + wire/doc_integrity_gate.sh as ready-to-wire gate scripts
- gates/ — only 3 files present (cm_covenant_114_162_propagation.sh, cm_opendesign_ui_system.sh + mutation test) — **no gate scripts yet exist here for §11.4.176/182/183/184/185/186/12.12** (those gates are “recommended” in the anchor text but not yet scaffolded as standalone files in gates/; §11.4.184/185/186 compliance is verified via the sonarqube/doc_integrity tool’s own exit codes + a propagation grep, not a gates/ file)
- hooks/ — guard-forbidden-commands.sh (§11.4.109), guard-track-branch-label.sh (§11.4.182), action_prefix_expand.sh, post-merge, test_guard_track_branch_label.sh
- multitrack/ — full toolkit: multitrack_claim.sh, multitrack_device_lock.sh, multitrack_registry.sh, multitrack_config.sh, multitrack_build.sh, multitrack-up/multitrack-down, track_label_audit.sh, etc. (implements §11.4.176/177/178/179/180/181/182)
- scheduled-work-engine/ — Go engine (cmd/, internal/, test/), DESIGN.md
- token_accounting.sh, subagent_tier.sh, enable_prompt_caching_check.sh — §11.4.141 token-efficiency helpers
- constitution/submodules/token_optimizer/ — Go module (pkg/, helix-deps.yaml) — depth-1 reusable engine
- constitution/submodules/session_orchestrator/ — Go module (alias/, claim/, helix-deps.yaml) — depth-1 reusable engine, session/alias/claim primitives underlying multitrack_*

Verified working (ran the real check script, not assumed):

```
$ bash constitution/scripts/sonarqube/sonarqube_install_check.sh
PASS scanner: /home/milos/Factory/software/sonar-scanner/bin/
sonar-scanner – SonarScanner CLI 8.0.1.6346
PASS podman: podman version 5.7.1
PASS compose runtime: podman-compose
PASS curl: present
PASS host vm.max_map_count=2147483642 (ES-ready)
=== install check exit 0 ===
```

What THIS project currently wires — verified, not assumed

Item	Status	Evidence
.claude/settings.json (repo root)	MISSING	ls .claude/settings.json → no such file
§11.4.109 PreToolUse guard-forbidden- commands.sh hook	NOT WIRED	No settings file exists to register it; not present in ~/.claude/ settings.json either (global settings only has a UserPromptSubmit → codegraph prompt-hook)
§11.4.182 PreToolUse guard-track-branch- label.sh hook	NOT WIRED	Same — no hook registration anywhere (project or global)
sonar-scanner on PATH	YES	which sonar-scanner → /home/ milos/Factory/software/sonar- scanner/bin/sonar-scanner; durable export found in ~/.bashrc:287-288 tagged §11.4.184 – durable sonar- scanner PATH (ATMOSphere, added 2026-07-09)
sonarqube_install_check.sh	PASSES (exit 0)	Ran it live — see output above
sonar-project.properties (project scan config)	MISSING	find . -iname sonar- project.properties → none. Not required by §11.4.184 (tooling- availability mandate, not scan-

Item	Status	Evidence
docs/AGENT_GUARDRAILS.md (project-level, per §11.4.109)	MISSING	on-every-build), but needed if/ when a scan is actually run Only exists at constitution/docs/ AGENT_GUARDRAILS.md — the consuming project has never created its own copy/pointer doc
Project carrier files' highest anchor (CLAUDE.md/ AGENTS.md/GEMINI.md)	STALLED at §11.4.166	grep -oE '§11\.4\.[0-9]+' CLAUDE.md \ sort -u \ tail → highest is §11.4.166; same for AGENTS.md and GEMINI.md. Anchors §11.4.167 through §11.4.186 (20 anchors) + §12.12 are absent from all three project carrier files
QWEN.md	MISSING ENTIRELY	ls QWEN.md → no such file (violates the four-carrier fleet pattern §11.4.157 establishes: CLAUDE/AGENTS/QWEN/GEMINI)
tests/ pre_build_verification.sh propagation gates	STOPS at §11.4.166	The script literally has run_gate "CM-COVENANT-114-153- PROPAGATION" ... "11.4.153" through ...-166-...; no gate lines for 167–186 or 12.12 exist
docs/workable_items.db (§11.4.93/.95 SSoT)	PRESENT (45KB)	This project DOES maintain multiple tracked-data representations (DB + Issues.md + Fixed.md + summaries + exports) — §11.4.186 squarely applies
.{consumer}/doc_integrity/ checkset.yaml	MISSING	No consumer checkset registered anywhere in the project tree
doc_integrity gate wired into export/sync/commit seams	NOT WIRED	No reference to doc_integrity/ doc-integrity outside constitution/
multitrack tooling in active use	NOT WIRED / LIKELY N/A	No project-level reference to multitrack_* outside constitution/; this project does not appear to run multiple concurrent git tracks/worktrees today, so §11.4.176/177/178/179/180/181/182 claim-registry + device-lock

Item	Status	Evidence
docs/features/Status.md (§11.4.153)	PRESENT	machinery is latent, not yet triggered — but the label convention (§11.4.182) still applies to every subagent dispatch regardless of track count Already exists — good, unrelated pre-existing compliance
scheduled-work-engine / token_optimizer / session_orchestrator	NOT CONSUMED	These are constitution-internal depth-1 engines (each carries its own helix-deps.yaml); no project-level integration point references them. Given this project doesn't run multi-track parallel work today, these are lower priority than the propagation/hook/doc-integrity gaps

Prioritized adoption checklist

P0 — Release blockers (mechanical propagation + verifiable gaps)

- Bump the constitution submodule pointer.** `git -C constitution` is at `e60cbde2`; the parent tree still records `fc4e9b8e`. Per §11.4.26 step 7 this MUST be bumped in the same commit as any cascade work.
 - Command: `git add constitution && git commit` (via `scripts/commit_all.sh`, never `raw git commit` per project convention) after the propagation work below lands.
- Propagate anchors §11.4.167–§11.4.186 + §12.12 into all four carrier files** (`CLAUDE.md`, `AGENTS.md`, `GEMINI.md`, and the still-missing `QWEN.md`) — currently stalled at §11.4.166. Required by §11.4.35/§11.4.157 (`GEMINI.md` lockstep — extends to the whole carrier fleet) and directly gated by `CM-COVENANT-114-1XX-PROPAGATION` for every one of the 20 new anchors.
 - Concrete gap: `tests/pre_build_verification.sh` has explicit `run_gate "CM-COVENANT-114-153-PROPAGATION" ... "-166-"` lines

and nothing beyond. Add the missing `run_gate` lines for 167–186 + 12-12, and restate each anchor’s literal token in the four carrier files.

3. **Create QWEN.md** at the project root — it does not exist at all. Every propagation gate pattern in the new anchors (CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep) assumes its existence.
4. **Wire the §11.4.109 PreToolUse guard hook** (`constitution/scripts/hooks/guard-forbidden-commands.sh`) into a project-level `.claude/settings.json` — the file does not exist at all today, so this hook (mandatory since an earlier round, not just this batch) has never been mechanically enforced for this project.
 - Command/config: create `.claude/settings.json` with a PreToolUse hook entry invoking `constitution/scripts/hooks/guard-forbidden-commands.sh` (inherited by reference, never copied, per §11.4.177).
5. **Wire the §11.4.182 PreToolUse guard hook** (`constitution/scripts/hooks/guard-track-branch-label.sh`) into the same `.claude/settings.json`, blocking any Agent/Task dispatch whose description doesn’t start with `^(T[0-9]+/[^)]+)`.
 - Note: even for a single-track project, `<N>` and `<branch>` resolve honestly (`T1/current branch`) or ? per §11.4.6 — the label convention still applies to every subagent dispatch this project makes.
6. **Create docs/AGENT_GUARDRAILS.md** at the project level (currently only exists inside `constitution/docs/`) containing the SUBAGENT CONSTITUTIONAL PREAMBLE + ORCHESTRATOR PRE-ACTION CHECKLIST sections the orchestrator pastes into every subagent dispatch, per §11.4.109.
7. **Wire the §11.4.186 doc-integrity anti-divergence gate** — this project maintains ≥ 2 representations of the same tracked data (`docs/workable_items.db` + `Issues.md/Fixed.md` + their summaries + `md/html/pdf/docx` exports), so §11.4.186 squarely applies and is currently completely unwired.
 - Steps: (a) go `build -o doc-integrity ./cmd/doc_integrity` from `constitution/scripts/doc_integrity/`; (b) create `.helix_ota/doc_integrity/checkset.yaml` registering the DB + `Issues/Fixed` + summaries as sources (schema in `constitution/scripts/`

doc_integrity/README.md + DESIGN.md §1.4); (c) wire doc-integrity verify as a blocking pre-step before scripts/export_docs.sh, before any sync_issues_docs-class regeneration, and before any doc-set commit; (d) run doc-integrity selfcheck to prove the golden-good/golden-bad/negative-control fixtures pass (§11.4.107(10) self-validation) before trusting the gate.

P1 — Should-wire soon (concrete but lower immediate blast radius)

1. Add SonarQube scan config + wire into tests/

pre_build_verification.sh. The CLI is installed and PATH-durable (§11.4.184 tooling-availability requirement is already SATISFIED), but there is no sonar-project.properties and no gate invoking sonarqube_install_check.sh from the project's own verification pipeline. Add a run_gate "CM-SONARQUBE-CLI-INSTALLED" bash constitution/scripts/sonarqube/sonarqube_install_check.sh line, plus a sonar-project.properties at repo root (project key helix_ota, sources server/, submodules/) if/when an actual scan (not just tooling-presence) is desired.

2. Document the §11.4.183 maximal-multi-agent-utilization

discipline explicitly in the project's own working-mode notes (e.g. docs/CONTINUATION.md or a project-specific addendum) — it's a discipline anchor with no standalone tooling to wire, but the propagation gate still needs the literal 11.4.183 present in the carrier files (covered by item 2 above) and the *practice* (parallel subagent dispatch, independent review agents) should be visibly followed in session logs.

3. Add a §11.4.185 manual-QA-final-confirmation checkpoint

to the release/tag workflow docs (docs/procedures/issues/Resolution.md or the release-tag script) — currently the release-tag terminal condition (§11.4.126) has no explicit “hand off to QA team, wait for manual confirmation” step documented anywhere in this project's own procedures.

4. Add a §12.12 thread-headroom pre-flight check

before dispatching >2-3 parallel subagents — e.g. a one-line ulimit -u + ps -L -u "\$USER" | wc -l check in whatever pre-flight/pre-dispatch helper this project uses (none currently exists; could be a small addition to scripts/ or referenced inline in docs/ AGENT_GUARDRAILS.md once created per item 6).

P2 — Latent / lower priority until the project actually needs them

1. **Multitrack tooling (§11.4.176–§11.4.181)** — not currently applicable since this project does not appear to run ≥ 2 concurrent git tracks/worktrees on shared resources today. Wire constitution/scripts/multitrack/* (by reference, per §11.4.177 — never copy) only when/if parallel-track work on this project actually starts. Until then, only the §11.4.182 label-prefix convention (item 5, P0) needs to apply.
 2. **scheduled-work-engine / token_optimizer / session_orchestrator submodules** — these are constitution-internal depth-1 Go engines with their own helix-deps.yaml. No project-level consumer exists. Evaluate for adoption only if this project adopts scheduled autonomous-loop work queuing or wants the token-efficiency (§11.4.141) tooling (token_accounting.sh/subagent_tier.sh/enable_prompt_caching_check.sh) wired into its own session workflow — none of those are wired today either, but §11.4.141 predates this 59-commit batch and is out of this checklist's primary scope.
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Honest boundary (§11.4.6)

This checklist proves what is/isn't wired via direct file reads, which, grep, and running the real sonarqube_install_check.sh script — it does NOT itself constitute the fixes. Every P0/P1 item above still needs its own §11.4.43 RED → GREEN implementation, §11.4.125/§11.4.142 independent review, and captured evidence before being marked closed.